

Multiple Choice (10 marks)

1. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
2. In Python 3, let $r = \text{lambda } q: q * 2$. What is $r(3)$?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
3. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above
4. In Python 3, what is the output of `print list[1:3]` if `list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]`?
`786 , 2.23, 'john', 70.2`
 - (a) `abcd`
 - (b) `786, 2.23`
 - (c) `None of the above.`
5. Let $x = 8$. What is $x >> 1$ in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8

True / False (10 marks)

Answer True or False

6. True or False: The output of the expression `4 * 1 ** 4` in Python is 4.
7. True or False: The condition `a < b` is sufficient to ensure that the expression evaluates to true.
8. True or False: Python supports automatic garbage collection, which helps manage memory efficiently.
9. True or False: In Python 3, accessing `'abc'[-1]` results in an error due to an invalid index.
10. True or False: In Python 3, the expression `b[::2]` returns a list containing `[11, 15, 19]` from the original list `b`.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the selection sort algorithm operates, focusing on the significance of comparisons in relation to the initial arrangement of elements in the list.
12. Discuss the characteristics of phishing attacks and analyze why an email from your bank asking you to call the number on your card is less likely to be a phishing attempt.

End of Paper